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REVIEW ARTICLE

Ground effect on rotorcraft unmanned aerial vehicles: a review

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Abstract
This article aims at collecting and discussing the results reached by the research community regarding the study of the ground effect on small rotorcraft unmanned aerial vehicles, especially from the modeling and control point of view. Rotorcraft performance is affected by the presence of the ground or any other boundary that alters the flow into the rotors. Specifically, the ground effect can induce perturbations in the flight stability, when operating near the ground. For a rotorcraft, an accident is likely to happen when the vehicle leaves or enters the ground effect region, which may cause crashes and property damages. Today, the use of unmanned aerial vehicles has grown widespread, which raises safety concerns when they are flying at very low altitudes and near the ground. Consequently, studying the influence of the ground over rotorcrafts is of paramount importance for general safety. Also, these investigations can be used to design systems of guidance, navigation, and control. In this review, we break down the most relevant works to date. We discuss aspects related to modeling, control, and application of the ground effect for small-scale multirotors, as well as other aerodynamic proximity effects, such as the ceiling and wall effects. We conclude by mentioning potential avenues of research when studying the ground effect from the point of view of the robotics and artificial intelligence fields.

Keywords Micro-air vehicle · Unmanned aerial vehicle · Rotorcraft · Rotary-wing aircraft · Ground effect

1 Introduction

Today, the use of small uncrewed rotorcraft, including helicopters and multirotors, has grown widespread in applications such as search and rescue [1], remote inspection [2], and aerial videography [3]. Furthermore, there are growing interests in using multirotors for transportation and delivery services [4,5].

On the other hand, the operation of these aerial vehicles at low altitudes and near the ground raises safety concerns because an accident is likely to happen when a rotorcraft leaves or enters the ground effect (GE) region, which may cause crashes and property damages. Specifically, the GE is a condition of improved performance when operating near the ground. Many helicopter accidents have occurred due to this aerodynamic effect, both non-fatal and fatal [6,7]. Consequently, studying the influence of the ground over rotorcrafts of any size is important for general safety.

For rotorcraft unmanned aerial vehicles (UAVs), investigations of the GE can be used to design systems of guidance, navigation, and control. For example, models of GE can be exploited to design/select control schemes [8,9]. At the path planning stage, predictions of the GE can be used to select paths that are less likely to suffer disturbances or to supply safety margins. Alternatively, predictions can be used to deliberately select paths that provide some impulse, thus reducing energy consumption [10]. At the control level, a model of the GE can be incorporated into the control loop, allowing to track trajectories more accurately [11,12]. Such algorithms are expected to improve well-known flight missions like hovering, picking-up and releasing objects, taking-off and landing in hard conditions, and operating near the ground in general.

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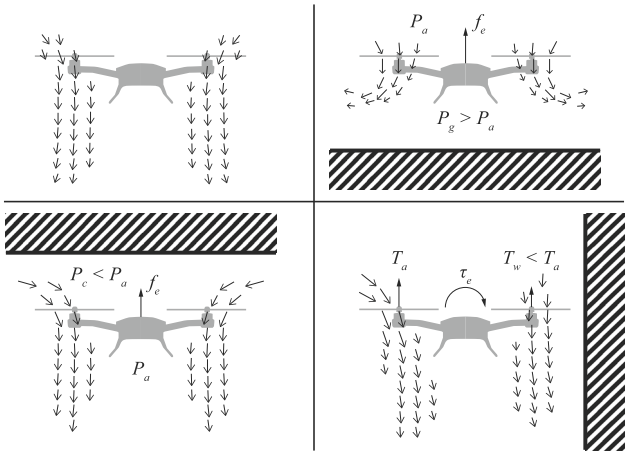


Fig. 1 Qualitative schematic of flow vectors out of proximity effects, in ground effect, in ceiling effect, and in wall effect. Pressure is denoted as P, thrust as T, force as f, and torque as τ. Subscripts a, g, c, w, and e indicate away from surface, near ground, near ceiling, near wall, and external, respectively

The study of the effect of the ground over rotorcrafts can be traced back to the 1930s. Around that time, it was already known that the wake of a propeller near the ground is deflected outward radially. Before 1980, a few analytical approximations of the thrust and power were proposed for a single rotor disk parallel to the ground [13–15]. Also, several experimental studies had been carried out for full-scale helicopters in GE [16], and there was an emerging interest in evaluating the impact of the ground on tilt rotors [17]. It is important to review previous work on helicopters since some of them have inspired what follows. In the 1990s, academic research on small-scale rotorcraft (UAVs) began with projects aiming for basic autonomy [18]. The following decade saw the maturity of computational fluid dynamics (CFD) for predicting rotor performance in GE, and the first advancements in the control of model helicopters and multirotors operating in GE [8,9]. Eventually, research centered on the modeling, control, and application of the GE for small-scale multirotors, as well as on studying other proximity effects, such as the ceiling and wall effects. Refer to Fig. 1 for a diagram depicting the flow vectors around multirotors when in ground, ceiling, and wall effects.

To the best of the authors' knowledge, the present article is the first survey about the effect of the ground on small-scale rotorcrafts. This survey leads us to identify several problems, trends, and challenges (see Fig. 2), from which the challenge of *autonomous navigation in unknown confined environments* stands out. We argue that this challenge will benefit from artificial intelligence, computer vision, and control concepts. The remainder of the

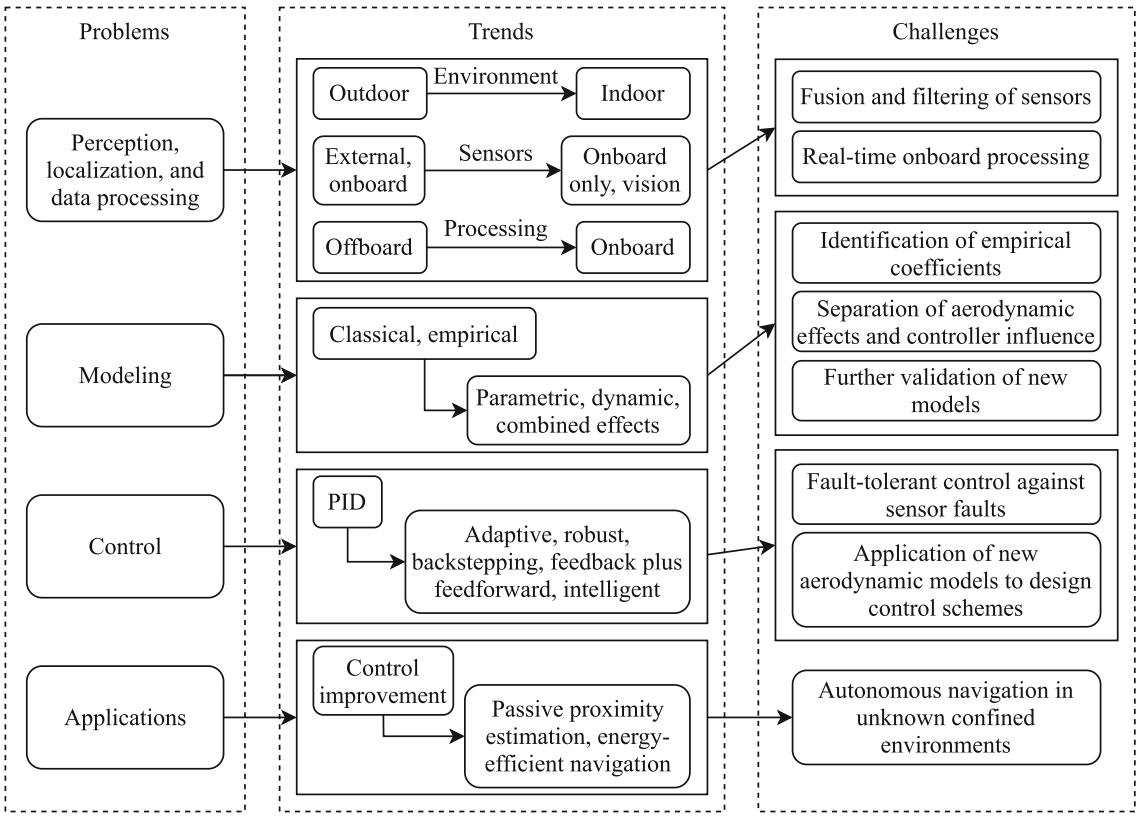


Fig. 2 Problems, trends, and challenges

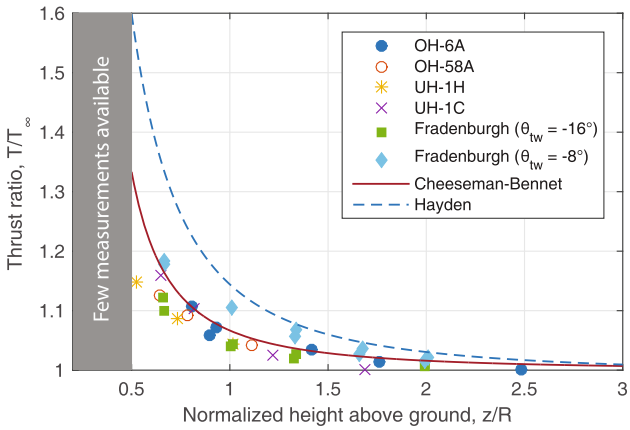


Fig. 3 Increase in rotor thrust versus distance from the ground for a variety of helicopters [19]. The legends OH-6A, OH-58A, UH-1H, and UH-1C are codenames for full-scale helicopters

article is organized as follows. Studies related to small-scale helicopters and multirotors are reviewed in Sects. 2 and 3, respectively. Section 4 briefly describes other aerodynamic proximity effects. Trends and challenges are discussed in Sect. 5 before the conclusion.

2 Background

In the study of the effect of the ground on full-scale helicopters, the use of scaled rotors is common. While the main focus in these studies is to generalize the results for full-size rotors, others have noted that small propellers behave fundamentally differently. This section discusses the research related to small rotors and model helicopters, but first, a summary of the topic's origin is given.

The effect of the ground on helicopters has long been recognized and studied. When a rotor is hovering in close proximity to the ground, the rotor slipstream is forced to expand radially as it approaches the surface. This alters the slipstream velocity, the induced velocity in the rotor, and, therefore, the rotor thrust and power [19]. The thrust of a hovering rotor in GE is found to be increased for a given power (or the required power is decreased for maintaining a given thrust). A representative plot of the thrust ratio in hover versus normalized altitude is shown in Fig. 3.

The influence of the ground can be viewed as a reduction of the rotor induced velocity, v_i, by a factor k_G [20,21], so that at constant thrust, the ratio of the induced power (or power coefficient) required to that out of GE is $C_P/C_{P\infty} = k_G$. Alternatively, the GE can be expressed in terms of the increase in thrust (or thrust coefficient) at constant power, which implies that $\lambda C_T = \lambda_\infty C_{T\infty}$ or $T/T_\infty = v_{i\infty}/v_i = 1/k_G$, where λ is the inflow ratio (induced velocity over rotor tip speed), and the infinity (∞) subscript indicates baseline values (far from the ground). Therefore, changes in thrust can also be interpreted as changes in v_i.

Literature about the GE on helicopters can be traced back to 1937. In that year, Betz [13] provided a theoretical estimate of the effect of the ground on the power of a propeller operating at constant thrust. In particular, he suggested that:

$$\left[\frac{P}{P_\infty}\right]_{T=\text{const}} = \frac{2z}{R} = k_G \quad z \ll R, \tag{1}$$

where z is the height off the ground, and R is the radius of the rotor. In 1941, Knight and Hefner [22], based on vortex theory, accounted for the ground by including image vortices below the ground plane. In 1947, Zbrozek [23] expressed the influence of the ground in terms of the thrust increase at constant power as a function of rotor height and blade loading. Cheeseman and Bennet [14], in 1955, sought to simplify the complex vortex interaction with a ground plane using momentum theory and the method of images, showing that the rotor thrust could be approximated by:

$$\left[\frac{T}{T_\infty}\right]_{P=\text{const}} = \frac{1}{1-(R/4z)^2}. \tag{2}$$

Equation (2) is limited because it does not consider viscosity or boundary layer effects and is only valid for $z/R > 0.5$. Despite this, it still fits well to helicopter flight test data. In 1960, Fradenburgh [16] conducted smoke flow experiments that showed the wake expansion in GE instead of the notional contraction observed out of GE. In 1976, Hayden [15] correlated flight test data to obtain the influence of the ground in hover. The profile part of the total power was assumed to be isolated from the induced effect, such that only the latter is influenced by the ground, that is $C_P = C_{P0} + k_G(C_{Pi})_\infty$, where:

$$k_G = \frac{1}{A + B(2R/z)^2}, \tag{3}$$

with $A = 0.9926$ and $B = 0.0379$. As shown in Fig. 3, Hayden's result slightly over-predict the rotor thrust. In all cases, it is apparent that the GE becomes negligible for rotors hovering greater than two rotor radii (one diameter) above the ground. Other studies have focused on the flows with vortex methods [24–26], grid-based CFD [27–29], vorticity transport models [30], and experimental data [31].

Apart from helicopters, the effect of the ground has been investigated for other vertical take-off and landing (VTOL) aircraft, such as tiltrotors (as the V-22 Osprey) and quad tilt rotors (as the V-44). Studies of the tiltrotor have shown that the vertical drag or download on the fuselage decreases as the height above ground decreases [17,32]. Moreover, it has been hypothesized that when the airflow from the rotors impacts

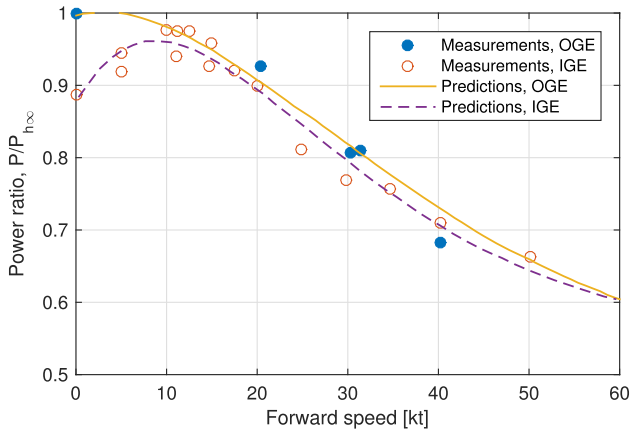


Fig. 4 Measurements and predictions of rotor power versus forward speed when operating near the ground [19]. IGE: in GE, OGE: out of GE

the ground, it forms a fountain flow below the center of the fuselage, resulting in a net upload [33]. This fountain flow also creates a net upload on the quad tilt rotor (QTR) fuselage [34]. It has been measured that the QTR thrust ratio reaches the out of GE value only after a height-to-radius ratio of 5 [35]. This is in contrast to tilt rotors where the thrust ratio reaches the out of GE value after a height-to-radius ratio of 3. For both the tilt rotor and the QTR, the effect of the ground fades away with the increase in height and forward velocity [36].

The effects of the ground on rotor performance in forward flight are also significant, but here the flow state tends to be more complicated. In forward flight, where the airflow is swept behind the rotor, the effect of the ground decreases quickly with forward speed. The GE is usually considered negligible for forward velocities greater than $2v_h$ or for advance ratios greater than 0.10, where v_h is the ideal hover induced velocity. A representative plot of the power ratio versus forward speed is shown in Fig. 4.

In the same report containing Eq. (2), Cheeseman and Bennet [14] presented a mathematical expression for the thrust ratio in forward flight:

$$\left[\frac{T}{T_\infty}\right]_{P=\text{const}} = \left[1 - \frac{(R/4z)^2}{1+(\mu/\lambda)^2}\right]^{-1}, \tag{4}$$

where μ is the advance ratio and λ is the inflow ratio. This formula displays the correct behavior as long as z/R is above 0.5. It should be observed that Eq. (2) is a particular case of Eq. (4), when $\mu = 0$. In addition, the effect of blade loading was incorporated into this model, but it will not be discussed here for the sake of brevity.

The GE on rotors in forward flight has been studied with analytical [37], experimental [38,39], and numerical techniques [25,27,40,41]. Similar to the hovering case, it has been observed that the ground reduces the power required, but the effect diminishes with height more rapidly in forward flight. On the other hand, for small heights and speed near hover, the net power required can actually increase because of the formation and influence of flow recirculation at the leading edge of the rotor [42]. At higher speeds, a well-defined horseshoe vortex is formed under the propeller, which becomes smaller and eventually vanishes as the advance ratio is increased.

To achieve real-time GE simulation, Khromov and Rand [43] presented a rotor wake model for computing the rotor inflow in combined forward and axial flight. Under the assumption of flat ground, they approximated the thrust ratio by the following expression:

$$\frac{T}{T_\infty} = e^{-Gz/R}(1+\bar{\mu}^2)^{-3/2} + 1, \tag{5}$$

where $\bar{\mu}$ is the advance ratio normalized by $v_h/\Omega R$, Ω is the rotor angular velocity, and G is a parameter related to blade loading and rotor inflow ratio, which is the ratio of v_i to rotor tip speed. Then, the authors expressed v_i by piecewise functions for all flight regimes, taking the axial velocity as input. This model was shown to have a reasonable correlation with hovering empirical data and forward flight results from the Cheeseman–Bennet model.

Acknowledging the influence of arbitrary surroundings, Bludau et al. [44] presented a real-time model to predict the rotorcraft motion in the vicinity of objects. They coupled a low-resolution fluid dynamics solver (based on the Lattice Boltzmann method) with a flight dynamics code (based on blade element momentum theory). Results from the coupled model were found to agree well with reference models (Cheeseman–Bennet, and Khromov), but guarantees were not given on its numerical convergence.

2.1 Control

Several control techniques have been proposed for small helicopters operating near ground, including adaptive backstepping [8], linear-quadratic with integral term designed for a model identified experimentally [45], gain scheduling of proportional–integral–derivative (PID) control as a function of Eq. (3) [46], and sliding mode control plus feedforward compensation with an empirical model of the lift force [11].

Three variants of the backstepping design procedure have been presented for altitude control for helicopters, including robust [47], adaptive [48], and robust adaptive [49]. In these works, a particular model of the rotor thrust is used, in which the thrust depends on the reduction of the induced velocity due to the GE. The adaptive component of the controller estimates and compensates a parameter that is related to v_i. Simulation results demonstrated that the tracking performance is improved by the compensation.

In another series of publications, a ring-source flow model for helicopters was developed [50–53]. In addition, a nonlinear model of the altitude of a heave test stand was derived, which facilitated the design of a linear quadratic regulator. The height was estimated from differential pressure readings using a Bayesian filter along with the flow model. Results from simulations and experiments indicated that height estimation and control is possible with only flow sensing and modeling.

2.2 Flow Studies

Research has been carried out about proximity flow effects on small rotors. Such rotors operate at significantly lower Reynolds numbers [19], which means that full-scale rotor studies might have limited applicability to the former. This claim motivated a work that used CFD to investigate the fluid interactions between a hovering small rotor and planar surfaces [54]. The computational method was validated by comparison against experimental results, a two-dimensional free vortex model, and results from existing literature. This study supposed that the variation in thrust generates a vertical force disturbance, $\bar{Z}$, acting on the helicopter. Then, the relationship between $\bar{Z}$ and ground proximity was characterized as:

$$\bar{Z} = mg \cdot a \left(\frac{z}{R}\right)^b, \tag{6}$$

with a and b fitted to the CFD results using least squares regression. Also, a linear simultaneous state and disturbance estimator was postulated. Flight tests showed that the scheme can provide fair in-flight estimates of disturbances that are induced by the ground plane using only an onboard inertial measurement unit (IMU). Furthermore, given that the relationship between proximity and disturbance magnitude is known a priori, it was possible to use the disturbance estimate to passively estimate proximity.

Prothin et al. [55–57] studied the influence of the surroundings (ground and wall) on the rotor's aerodynamic performance and the flow field. The results were analyzed and compared to the free flight case. The analysis was carried out by both numerical and experimental approaches, validating the first with results from the second. They disclosed that while the study revealed interesting features, further experiments and simulations are required.

2.3 Modeling

The modeling of the GE for small rotors has attracted the attention of the UAV community. Hosseini et al. [58] performed a study on a tilted ducted fan to characterize the GE while hovering and develop a practical model for real-time

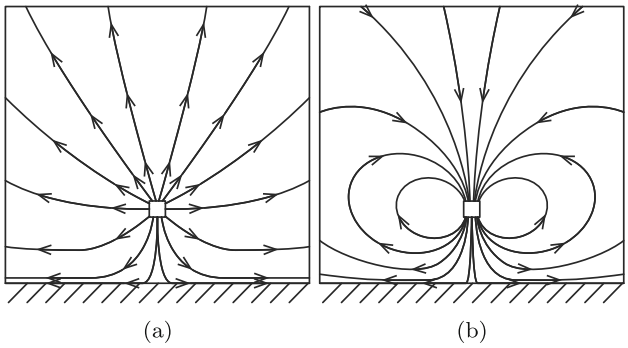


Fig. 5 The flow patterns for a fan in GE represented as **a** a source and **b** a dipole [58]

predictions. They performed CFD simulations, which were validated by empirical and theoretical information. The practical model was proposed as a dipole flow model (see Fig. 5) and was compared with the CFD results. It was shown that the dipole model, whose parameters were defined empirically, is able to predict the fan performance accurately.

Singularities exist in widely used empirical models (Cheeseman–Bennet and Hayden). Furthermore, blade geometry parameters are generally neglected. To overcome these drawbacks, a new GE model, inspired from Eq. (5), was proposed that predicts a finite maximum thrust ratio based on blade element theory, which was used to describe the model parameters in terms of the rotor pitch angle and solidity [59]. Specifically, the model is expressed as:

$$\frac{T}{T_\infty} = C_a e^{-C_b z/R} + 1, \tag{7}$$

where C_a and C_b are coefficients that depend on the blade geometry. This model is valid for the hovering case only. Experiments were carried out to validate the model. These results showed that the theoretical prediction provides a statistically significant relation to the experimental data.

3 Multirotor

Even though akin, helicopters and multirotors differ greatly on their mechanical structure and operation. In short, a helicopter is operated by varying the pitch angle of the main rotor blades and of the tail rotor blades. On the other hand, multirotors are equipped with fixed pitch propellers whose only operational variable is rotational speed. Moreover, the airflow around multiple lifting rotors is not the same as around a single lifting rotor. These differences have been the motivation of a research niche considering all aspects of small multirotors, including the GE.

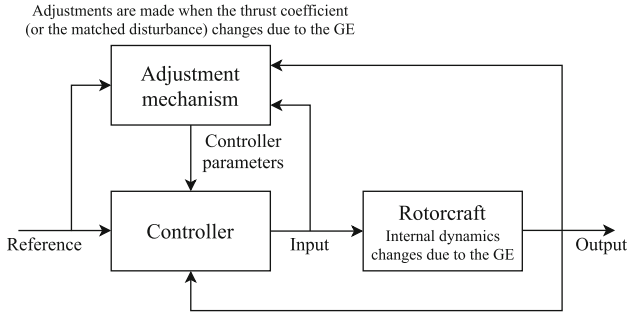


Fig. 6 Block diagram of the adaptive control strategy

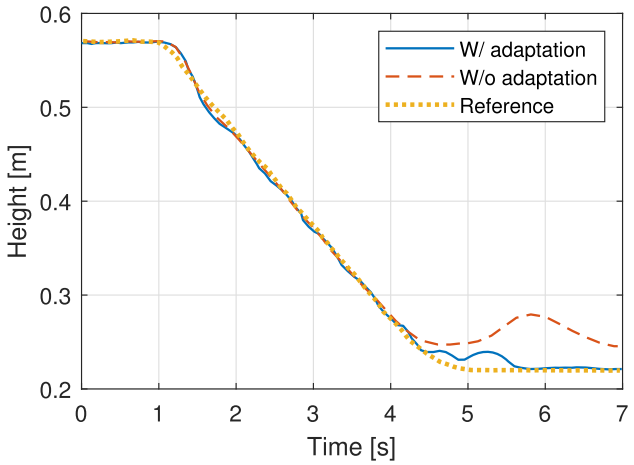


Fig. 7 Landing trajectory under adaptation and without adaptation [61]. Without adaptation, the quadrotor deviates greatly from the reference when it approaches to the ground; this is caused by the GE. When the quadrotor lands with adaptation, the error is significantly smaller

3.1 Control

As with model helicopters, various control approaches have been presented for multirotors operating in close proximity to the ground. For the problems that arise in this condition, the most widely used control strategy is adaptive control. The general structure of this strategy is depicted in Fig. 6.

Guenard et al. [9] proposed an adaptive altitude controller that estimates online the global thrust coefficient of a quadcopter (assuming that the thrust coefficient is the same for all the rotors). Particularly, the control law, u, was proposed to be as follows:

$$u = mg - mk_1(k_1 + k_2)\delta_2, \tag{8}$$

where k_1 and k_2 are fixed control parameters and δ_2 is the adaptable control parameter that is changed when the estimation of the thrust coefficient changes due to the GE. To deal with the noised value of the ultrasonic sensor used in the onboard altitude loop, they designed an estimator of the vertical position and velocity. It was noted that using the adaptive controller and the estimator worked well in experiments with a quadrotor prototype (only results for heights above 0.5 m were reported, situation in which the GE might not be significant given the size of the prototype), although bad measurements from the ultrasonic sensor can destabilize the system.

Lee et al. [60] compared in simulations two nonlinear control methods for quadcopters: feedback linearization and adaptive sliding mode. The first was designed as an x–y–z control, and the second as a trajectory tracking control. They considered an unknown acceleration term, representing the GE, while designing of the adaptive controller. In the tests, the GE term was assumed to be given by:

$$g_r(z) = \begin{cases} \frac{A}{(z+z_{cg})^2} - \frac{A}{(z_0+z_{cg})^2} & 0 < z < z_0 \\ 0 & \text{otherwise} \end{cases}, \tag{9}$$

where A is a constant, z_{cg} is the z-component of the center of gravity, and z_0 is the altitude under which the ground affects the vehicle. The authors designed an augmented control input, U, of the form:

$$U = G^{-1}(x)[F(x, x_d) + \hat{v} - \hat{f}_r(x)], \tag{10}$$

where x is the state, x_d is the desired state, $G(x)$ is the augmented input matrix, $F(x, x_d)$ is the control term with fixed parameters, $\hat{v}$ contains the estimation of the slack variables, and $\hat{f}_r(x)$ is the adaptable control term including the estimation of the GE force. It was found that the conventional feedback linearization controller is not robust to uncertainty and sensor noise. Also, it was noted that the adaptive sliding mode controller performs well under sensor noise and that it is able to compensate the uncertainty caused by the GE. In another paper by the same lead author, the adaptive part of a sliding mode controller was represented as a neural network whose weights were adjusted to approximate the values of additive unknown disturbances acting on the accelerations of the inertial z-axis (due to the GE), and of the body roll, pitch, and yaw angles. This adaptation algorithm was tested with a 0.8-kg quadcopter. The state of the vehicle was obtained from a Motion Capture (MoCap) system and an IMU. After control calculation in an offboard PC, the reference commands were sent to the quadrotor by a RC transmitter. Experimental results showed that, with the adaptation rule, the tracking performance was improved, especially along the z-axis.

Hu et al. [61] proposed a control structure to achieve the landing of a quadrotor onto a vertically oscillating platform. The structure consisted of three modules: a motion estimation module, a trajectory generation module, and an adaptive robust tracking controller. This last module was conceived with a simplified dynamics of the vertical direction. It estimates an uncertain equivalent inertia online, m/k_G, which is

altered by the GE. A theorem was provided on the boundness of the tracking error. Experiments with a Nano Plus quadrotor showcased that landing with adaptation reduces the tracking error when the quadrotor approaches the ground, as shown in Fig. 7. The controller was executed in an offboard computer using measurements of vertical position from a MoCap system, and vertical acceleration from an IMU.

A pair of authors have explored model reference adaptive control (MRAC) for quadcopters operating close to the ground. In the thesis of Arshad [62], a numerical formulation of MRAC was studied to compensate the GE. Simulations depicted that the addition of the adaptive controller improved the system response at the time of launch, provided that a PID controller was also present. However, the system response with adaptation is similar without adaptation when tracking a trajectory. It even induces more error when the vehicle approaches the ground. Wei et al. [63] presented a more effective MRAC scheme for mitigating the GE on mini quadcopters. First, a mathematical model was established through a set of experiments and validated by a flight test. It was found that the total thrust generated by the four rotors increases linearly as the multirotor approaches the ground when the rotor-to-rotor distance is small, and behaves as a quadratic function (similar to the Cheeseman–Bennet model) when the rotor to rotor distance is increased. The linear model has the form:

$$\frac{T}{T_\infty} = \begin{cases} a\frac{z}{R} + b & \text{if } \frac{z}{R} < c \\ 1 & \text{otherwise} \end{cases}, \tag{11}$$

where the values of a, b, and c are retrieved from empirical data. Then, the (linear-quadratic) attitude and position controllers were introduced. It was in the position loop that the MRAC was added to a traditional PID controller. The PID was designed as a 3D position tracker, and the MRAC was added to the z direction to counteract the GE. The idea is to consider a matched uncertainty (an uncertainty that enters the system through the same channel as the control input) in the original system, which can be written as a basis function with unknown coefficients. To cancel said uncertainty, the adaptive control input, u_{adp}, is calculated as follows:

$$u_{\text{adp}} = -\hat{\Theta}^T \cdot \Phi(x), \tag{12}$$

where $\hat{\Theta}$ is the estimated unknown constant coefficients for the matched uncertainty and $\Phi(x)$ is the basis function. Particularly, the rate of change of $\hat{\Theta}$ is designed to depend on the difference between the measured states and the reference model states. Two basis functions were considered in the study, a linear function, and a series of radial basis functions (RBFs). Using the error between the real states and the reference model states, an additive control term is calculated to cancel out the said uncertainty. Experiments were performed with a ground station and a Crazyflie 2.0. The attitude loop was run onboard at 500 Hz and the position loop on the ground station at 200 Hz. In detail, the position and yaw angle of the Crazyflie was measured by a MoCap system. The results showed that the traditional PID is outperformed by the inclusion of the MRAC. Additionally, it was shown that the MRAC with RBFs has better performance than the MRAC with a linear model at the expense of extra computational load.

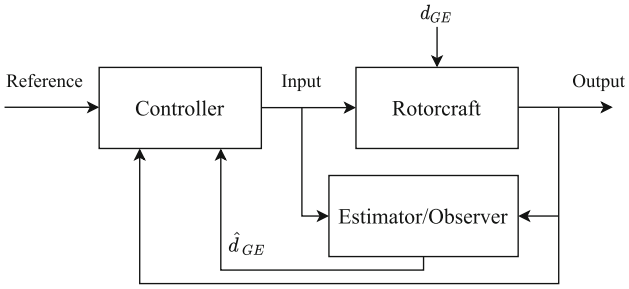


Fig. 8 Block diagram of the active disturbance rejection control strategy. The disturbance caused by the GE is denoted as d_{GE}, and the estimation the same disturbance as $\hat{d}_{\text{GE}}$

Control schemes based on estimators or observers are also common in the literature about multirotors in GE. Nobahari and Sharifi [64] implemented a continuous ant colony filter for the estimation of states and parameters. The filter requires a mathematical model of the process, which was developed for quadrotors. This model considers a GE coefficient inspired on the one proposed by Hayden, Eq. (3). Instead of using the radius of one rotor, they used a greater equivalent radius to compute the coefficient. In essence, this equivalent radius was introduced to magnify the effect of the ground for the quadcopter, an observation that has been noted in previous works. To deal with an inaccurate model of the GE, the filter was used to estimate and correct the errors online. However, the algorithm could not operate with a direct landing trajectory because, close to the end, it involves a near to hover condition, which is not observable in theory. For that reason, a special landing procedure was proposed containing a sinusoidal motion in the middle. This motion allows the algorithm to correctly estimate the modeling errors, which are consumed in the final phase. Simulation results are provided, showing that the automatic landing is accurately performed with the proposed procedure.

When the observer or estimator is used to estimate internal or external disturbances online in the control loop, the control strategy is often referred to as an active disturbance rejection control. Figure 8 outlines the general block diagram of this strategy. Du et al. [65] decoupled the quadrotor control system into three loops: position, angle, and inner loops. Dynamic inversion was applied to both the position and angle loops, while incremental nonlinear dynamic inver-

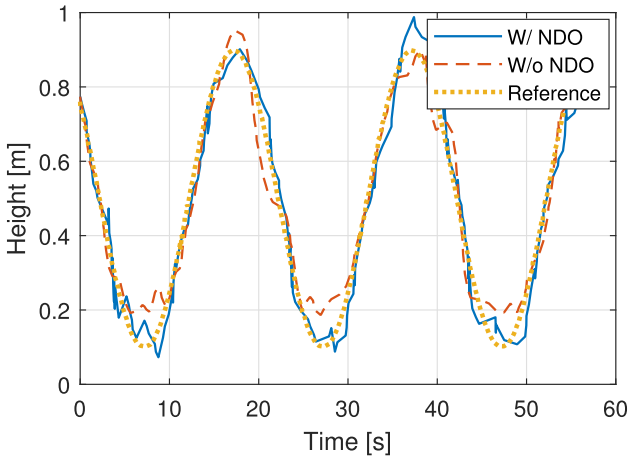


Fig. 9 Trajectory tracking under the nominal controller with NDO and without NDO [66]. The NDO allows compensating for the additional force, which is caused by the GE, on the quadrotor, and helps the quadrotor to reach the desired minimum height

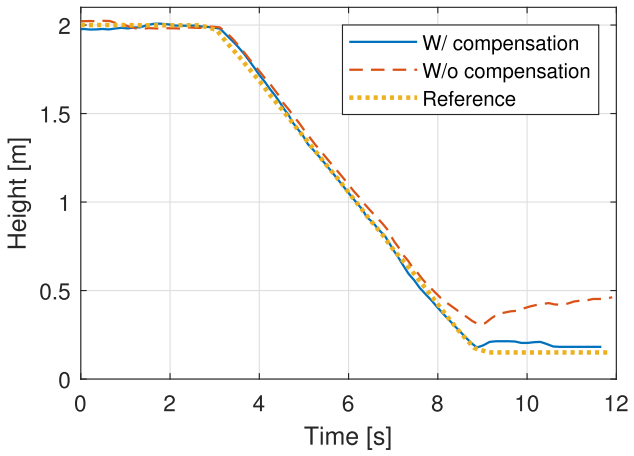


Fig. 10 Landing trajectory under the nominal controller with compensation (robust and GE) and without compensation [69]. Without the compensation, the quadrotor could not track well the reference when the GE is significant. A precise landing with accurate time can be achieved with the compensation

sion was applied to the inner loop. To achieve GE rejection, integral extended state observer (IESO) was combined with the inner loop. Besides, the arranged transient process technique is utilized to smooth out jumps in the reference. Similar to (10), the innermost control law is designed as:

$$u = w_0 + G^{-1}(x)[F(x, x_d) - \Delta_{\text{dist}}], \tag{13}$$

where w_0 refers to the velocities of the rotors in the last sample time and Δ_{dist} to the observed external disturbance accounting for the GE. The ability of IESO to observe and compensate the GE was verified only in simulations.

Similar to the previous work, He et al. [67] proposed to enhance the inner (attitude) loop with a nonlinear disturbance observer (NDO), while the outer (position) loop was commanded by a PID controller. The nominal controller of the inner loop consisted of a commonly used proportional–integral–velocity controller. For its part, the NDO was in charge of estimating the disturbance (force and torque) generated by the GE. This estimation is converted to PWM (pulse-width modulation) values and then added to the output of the motor mixer to compensate for the GE. To simplify the modeling process, an adaptive law was designed for the global thrust coefficient. The proposed control scheme was evaluated in simulations with the full quadcopter dynamics and the GE model in Eq. (7). In fact, the reduced version of this model was first described in this work. Using BET, the maximum GE ratio was found to be $\max(T/T_\infty) \approx 1.7$. Simulation results showed that the NDO can effectively compensate for the GE. In a subsequent publication [66], this control scheme, without the adaptation law, was validated in experiments. Some qualitative results are shown in Fig. 9. The NDO was implemented onboard a quadcopter and a MoCap system provided the position feedback.

As a side note, the height velocity, consumed by the inner loop, was estimated by fusing the pressure measurement from a barometer and the vertical acceleration from an IMU.

Robust controllers can be used to compensate for bounded disturbances and parametric uncertainties in nonlinear systems, such as multirotors. Waheed et al. [68] performed a numerical simulation of a robust controller applied to a quadrotor. In particular, they developed a position tracking controller using the saturated robust integral of the sign of error (RISE). The expression of the controller, u, is given by:

$$u = G^{-1}(x)[\gamma_1 \tanh v], \tag{14}$$

in which γ_1 is a positive constant gain and v is the Filippov solution of a differential equation containing specially designed filtered tracking errors. The GE was simulated in the same manner as in Eq. (9). The numerical outcomes showcased that the saturated RISE is able to utilize the benefits of high-gain control while assuring that the saturation limits are not exceeded.

Danjun et al. [69] opted for designing a robust altitude controller by combining a PID controller with a robust compensator. The control input, u_z, consists of two parts, the nominal (PID) input u_n, and the robust compensating input $u_{z,r}$, so that $u_z = u_{z,n} + u_{z,r}$. Explicitly, $u_{z,r}$ is provided as:

$$u_{z,r} = f_z g_z (z_{z2} - z)m, \tag{15}$$

where z is the position of the vehicle along the inertial vertical axis, m is the mass of the vehicle, and z_{z2} is obtained from a system of two ordinary differential equations resulting from applying a robust filter, with roots in f_z and g_z, to the z-axis

dynamics. Moreover, they presented the following GE model for quadrotors based on Eq. (2):

$$\frac{T_{\text{input}}}{T_{\text{output}}} = 1 - \rho \left(\frac{R}{4z}\right)^2, \tag{16}$$

where T_{input} is the given thrust command, T_{output} is the actual thrust generated by the quadrotor, and ρ is an empirical coefficient that needs to be determined with hover tests at different altitudes. This model was used to implement a GE compensator. In outdoor experiments, they showed that the inclusion of a well-adjusted GE compensator to the PID controller improves the tracking performance. On the other hand, the robust compensator provided little enhancements to the tracking performance. An extract of the qualitative results are plotted in Fig. 10. Also, the results indicated that the unmodified Cheeseman–Bennet model is not suitable for quadrotors. In the control loop, the altitude was measured with a camera and an ultrasonic sensor.

Keeping robustness in mind, Shi et al. presented a deep-learning-based nonlinear controller (dubbed as neural lander) that improves the performance of a quadcopter during landing [70]. Their approach combines a nominal dynamics model with an artificial neural network (ANN) that learns high-order interactions offline. Spectral normalization is applied to the ANN to bound its output. This bound is exploited to design a feedback linearization controller, which uses the learned model online, and prove system stability. The control input is of the form:

$$u = B^{-1}(u_{\text{PD}} - mg - \hat{f}_a), \tag{17}$$

in which B is the input matrix, u_{PD} is the output of a PD controller, g is the gravity vector, and $\hat{f}_a$ is the ANN prediction of the disturbance force induced by the GE. The overall control performance was evaluated in experiments with a MoCap system (for accurate position, attitude and velocity estimation at 100 Hz), a WiFi router for communication, and an Intel Aero drone. It was demonstrated that the proposed controller significantly outperforms a baseline nonlinear tracking controller in the cases of landing and cross-table trajectory tracking. Motivated by these results, From's thesis [71] reported the implementation of an ANN to predict the GE from the state and control input of a drone. As an extension, online learning was added to the ANN, with the final objective of adapting to new situations that are not known in advance. As in the neural lander, the ANN approximation of the GE was integrated into a nonlinear feedback-linearizing controller. The overall system was tested in simulations, in which the GE was represented with Eq. (16). It was found that the online network was able to slowly adapt to new ground effects and showed in general as good results as and even better than the offline-trained network. However, no stability guarantees were given for the closed-loop system.

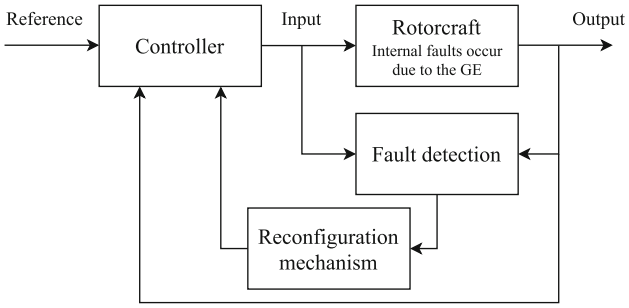


Fig. 11 Block diagram of the fault-tolerant control strategy

Some studies have innovated in the sensing process of VTOL vehicles for landing in moving platforms while considering the GE. This is the case of a paper by Herissé et al. [72], in which a nonlinear (outer loop) controller that exploits optical flow is described. They consider that the vertical dynamics has a slowly varying parameter that models the GE, $b \geq 1$ and $\max(b) = b_{\max}$ when $z = 0$. Next, the stability analysis of the landing controller, which regulates the flow divergence around zero, led to an inequality relating the gain of the controller and $b_{\max}$. This means, in essence, that the bound of the GE is taken into account in the design the controller gain. In experiments, video and inertial measurements from onboard sensors of an X4 flyer (quadrotor) are transmitted to a ground station, which computes the desired orientation and thrust that are sent to the drone's autopilot. It was shown that the stability condition is not always verified, which caused unwanted oscillations during the landing phase. This problem is mainly because of experimental constraints (large latency and sampling time). However, the UAV was able to land with satisfactory behavior. For the same problem, Lee et al. [73] used image-based visual servoing to track the platform (with fiducial markers) and generate a velocity reference command. This reference is used as the input of an adaptive sliding mode controller that compensates the GE. Measurements from a MoCap system, an onboard camera, and onboard IMU are sent to an offboard PC, which computes the commands that are sent to the vehicle's autopilot. Experimental evidence showed satisfactory landing behavior, in which the GE is estimated and compensated.

Besides the increment in thrust efficiency, the GE causes variations in the pressure altitude measured by barometers [75] and distance measurements of ultrasonic sensors [9]. Even though several control schemes have been proposed for compensating the GE, most of them have only considered the case in which there are no faults in the rotorcraft. While studying a multicontroller structure for rejecting the GE [76], Matus et al. noticed that the GE was inducing altime-

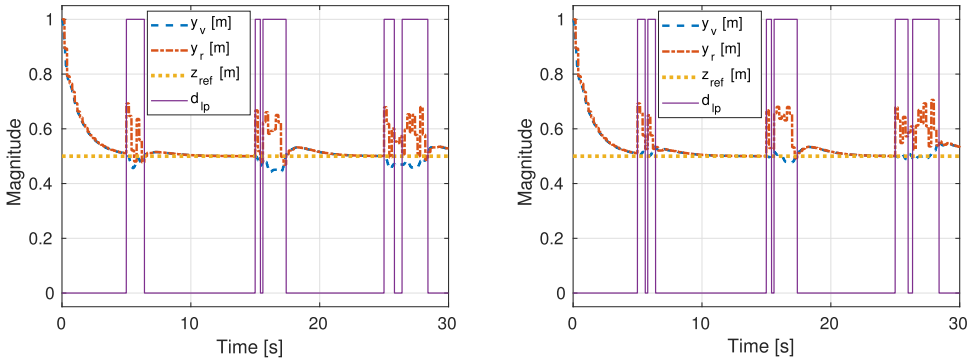


Fig. 12 Hovering behavior in GE for two controllers [74]. The first is a PD controller, and the second is a fault-tolerant controller, which is built on the first. The height measurement from vision, the height measurement from internal sensors, the reference, and the sensor fault detection are denoted as y_v, y_r, z_{ref}, and d_{lp}, respectively. It can be seen that the faults can be mitigated after being detected with the FTC strategy

ter sensor faults which degraded the control performance. Later, the authors developed a fault-tolerant control (FTC) strategy for multirotors with altimeter faults caused by the GE [74,77]. The general block diagram for this control strategy is outlined in Fig. 11. In particular, the authors assumed a hierarchical control structure consisting of an external (position) controller and an internal (velocity) controller. It was considered that sensor faults occur within the internal loop and the corresponding mitigation within the external loop. First, the sensor fault is detected with a fault diagnosis scheme, which was designed as a logical process depending on the weighted residual between the internal measurement and the external estimation from a metric monocular simultaneous localization and mapping (SLAM) algorithm. Once detected, the fault is mitigated using a function of the residual. This strategy was shown to be effective in simulations and experiments. Figure 12 presents an excerpt from the simulations.

For the benefit of the reader, the control strategies applied to the problem of the GE are summarized in Table 1.

3.2 Modeling

A number of models of the GE for multirotors have been presented already in the previous subsection. However, the conception of those models has not been the primary concern and little or no details have been given about their validity. In this subsection, we will discuss the research that revolves around the modeling of the effect of the ground on multirotors.

The earliest studies on GE modeling for multirotors were focused on applying or evaluating the classical models of thrust change obtained for helicopters. Bangura and Mahony [78] proposed a hybrid dynamic model to account for different flight states of the vehicle, including GE. The idea is to switch between states (normal, vortex ring, turbulent wake state, windmill brake state, and GE) given some predefined conditions. Equation (4) (Cheeseman–Bennet model for forward flight) was used to represent the GE. In a different report [79], the 4 in Eq. (2) (Cheeseman–Bennet model for hovering) was replaced with a constant that was later identified with empirical data collected for the kQuadNano. A similar procedure with the same quadrotor was reported by Powers et al. [80]. In their data, the GE can be observed up to $z/R = 5$, which is significantly higher than predicted by Eq. (2). Sharf et al. [81] assessed the applicability of this model for the Draganflyer X8, which is a small rotary-wing UAV with eight propellers arranged in four coaxial pairs. Again, it was found that the GE was apparent at higher altitudes than predicted by the model.

The rotors that are used in model helicopters are different from the ones used in small multirotors. This has motivated the study of the GE in the hovering performance of the latter. The work presented in [83] focused on developing a numerical method for predicting the thrust and power consumption in GE. The authors developed a model with blade element momentum theory which is combined with Cheeseman–Bennet predictions. The resulting model provides a purely analytical method for modeling the GE performance of a specific propeller geometry. However, the authors do not explore potential interactions between multiple adjacent rotors. An experimental study [84] assessed the applicability of Eq. (2) for a small hovering rotor (between 6 and 20 in). The results showed that this expression is a good approximation for small isolated propellers too. In a follow-up publication [82], a method was developed which allows the generation of a map of aerodynamic effects in an environment with multiple objects. First, the aerodynamic effects close to surfaces were experimentally tested along a two-dimensional grid (Fig. 13a). Then, the data are combined to account for different objects in the aerodynamic map (Fig. 13b). On the same line, Bernard et al. [85] performed an extensive experimental campaign aimed at characterizing the GE for a small quadcopter. In particular, both an isolated

Table 1 Control strategies presented in the literature

Control strategy	Sub-strategy	Remarks	Pros and Cons
Adaptive	Backstepping [8], measurement estimation [9], sliding mode [60], backstepping [48], robust [49,61], MRAC [62,63], ANN [71]	The objective of this strategy is to adapt the controller to the thrust coefficient or external force that changes owing to the presence of the ground	P: does not excessively rely on the model, does not trade performance for modeling accuracy; C: does have trouble adapting to arbitrary disturbances
LQR	Integral term [45]	Here, the state feedback gains are optimized using a predefined model of vehicle dynamics, which is linearized and takes into account the GE	P: computationally efficient (at runtime), optimal energy-wise; C: slow convergence, relies heavily on the model (at the design stage)
Gain scheduling	PID [46]	The gains are scheduled as a function of Eq. (2)	P: exploits linear control design methods; C: guarantees local stability only
Sliding mode	Integral term [11]	An empirical model is used to compensate for the GE and the sliding mode control to cope with the modeling error	P: robustness against matched disturbances and uncertainties; C: steady-state error when out of GE, chattering phenomenon
Observer/estimator-based	Probabilistic filter [64], IESO [65], NDO [67]	This strategy uses either an observer or an estimator for observing or estimating disturbances caused by the GE	P: great transient response, easy implementation (observer); C: overly sensitive to noise (observer), expensive computationally (estimator)
Robust	Backstepping [47], RISE [68], PID + compensator [69], ANN [70]	The ground effect is compensated with theoretical guarantees about the stability of the system	P: asymptotic convergence; C: trades performance for robustness
Fault-tolerant	Sensor fault-tolerant [74,77]	Sensor faults caused by the presence of the ground are diagnosed and then mitigated online	P: ability to deal with sensor faults; C: unnecessary without occurring faults
Non-conventional feedback	Optical flow [72], visual servoing [73], depth map [12], differential pressure [51]	The height above ground, which is the conventional feedback, is replaced with another measurement. Any of the above control strategies might be used	P: might exploit linear control design methods; C: employs uncommon sensors, might require preprocessing

rotor and the complete platform were studied. The collected results were compared against the Cheeseman–Bennet model and the model of Hayden, showing that the effect of the ground is extended up to almost $z/R = 4$. Conyers et al. [86] went a step further by reporting empirical data for T/T_∞ from two different spacings between propellers on a typical x-shaped quadrotor frame. The results suggest that Eq. (2) cannot accurately predict GE performance for quadrotors. In his thesis, Conyers [87] presented more empirical data for different propeller sizes and propeller spacings. These data were used to fit a rational polynomial for T/T_∞ as a function of normalized height $x = z/R$, propeller spacing L, and propeller radius. A compromise was made to use an equation with the lowest order while still producing a quality fit. The final equation form is defined by:

$$\frac{T}{T_\infty} = \frac{p_1 x^4 + p_2 x^3 + p_3 x^2 + p_4 x + p_5}{x^3 + q_1 x^2 + q_2 x + q_3}, \tag{18}$$

where p_i $i = 1, 2, \ldots, 5$ and q_j $j = 1, 2, 3$ are defined as follows:

$$\begin{aligned} p_i(L, R) &= a_{i,1}L^2 + a_{i,2}L + a_{i,3}R^2 + a_{i,4}R + a_{i,5} \\ q_j(L, R) &= b_{j,1}L^2 + b_{j,2}L + b_{j,3}R^2 + b_{j,4}R + b_{j,5} \end{aligned}, \tag{19}$$

with constant coefficients $a_{i,i}$ and $b_{j,i}$ that are obtained from the fitting process.

To prevent potential issues such as selection bias or overfitting, all coefficients in Eq. (19) were obtained from a cross-validation procedure, from which the mean accuracy of the model was found to be within the generally adequate level of ±5%. However, the author did not provide comparisons between the model output and measurements obtained under different conditions (radius and spacing) from the ones used in the cross-validation procedure.

Experience-based models have been developed for capturing the GE. Ryan and Kim [88] demonstrated how a vision

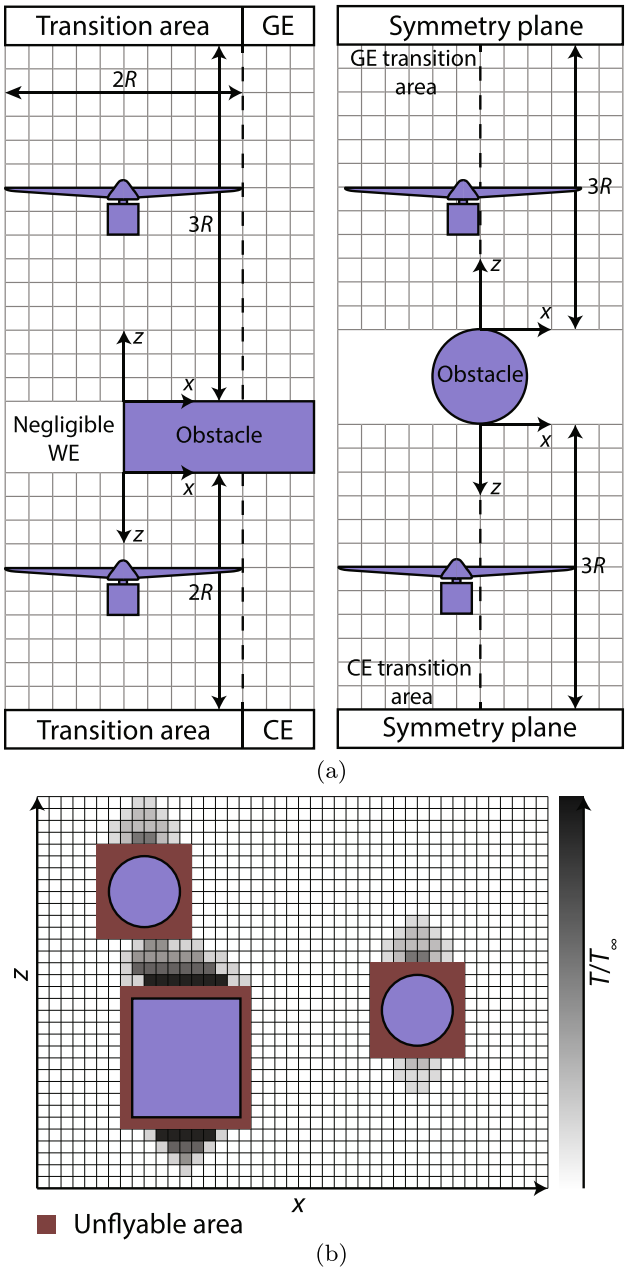


Fig. 13 Modeling and mapping of aerodynamic effects acting on a single rotor [82]

system can be used to provide information about the GE using environmental cues. The main idea is that streamers attached to the ground will move around faster due to the pressurized airflow causing the GE. Then, the ground cushion force was learned with an ϵ-support vector regression model using empirical data. Another study demonstrated that the modular recurrent neural network framework is capable of modeling a multi-input-multi-output model of a quadrotor in the presence of GE [89]. The GE can also be acquired through data-driven or low-dimensional stochastic models [90]. Karydis and Hsieh [91] derived a data-driven model based on principal orthogonal decomposition to capture the GE for a hovering quadcopter. This model closely predicts the sample mean of the testing dataset. Karydis et al. [92] proposed a systematic approach that extends a deterministic model to a stochastic regime. With this, they developed a reduced stochastic model for a quadrotor which captures the mean and variance of the empirical data at all setpoints in steady state.

In the classical rotorcraft literature, the GE is studied in static conditions, in essence, for a fixed rotor. Even when a non-fixed rotorcraft is analyzed, the data are collected once the vehicle is at a stable hovering state. The effect of distance from the ground on the dynamics of the attitude motion is ignored. This was the main concern of Aich et al. [93] who reported a frequency-domain analysis of the roll dynamics as a function of height for a non-fixed quadcopter. A slow-down of attitude dynamics as the distance from the ground decreases is registered. Closely related, Del Cont Bernard et al. [94] identified second-order models of the pitch dynamics of a quadrotor (operated on a single degree of freedom testbed, representative of actual near hovering condition). These models show a dependence with respect to the height from the ground, justifying the degradation of the attitude control effectiveness in near-ground flight.

When a rotorcraft flies past over an object, it is disturbed by the GE. Bartholomew et al. [95] proposed a system for predicting the acceleration expected when a quadcopter flies over an object. They use Gaussian process regression to learn a function that receives a depth image and yields a vertical acceleration. The qualitative results show that the predictor output is similar to the ground truth for several seen objects and just one unseen object. Also, the authors present quantitative results (average error), which they used only to compare the error for the same obstacle in different data-sets. The quantitative results provide the error mean for four unseen objects. Since the approach does not consider the velocity of the drone, it is limited to predict accelerations which are valid for a small vicinity around the training flight velocity. In a later publication [12], the same authors incorporated the prediction system into the flight control loop. To target the undisrupted level (forward) flight, they use the predictions to adjust the throttle command when the drone passes over an object. The results display significant improvements for the box and step objects. On the other hand, the system was not able to compensate for the disturbance resulting from one already seen object and overcompensated for two unseen objects, even though they were similar to the box obstacle.

As discussed before, experimental results with small quadcopters suggest that the GE may be larger than predicted by Eq. (2). However, the issue had not been further analyzed. Sanchez-Cuevas et al. [96] provided an expression of the thrust increment due to the GE for coplanar quadrotors, which

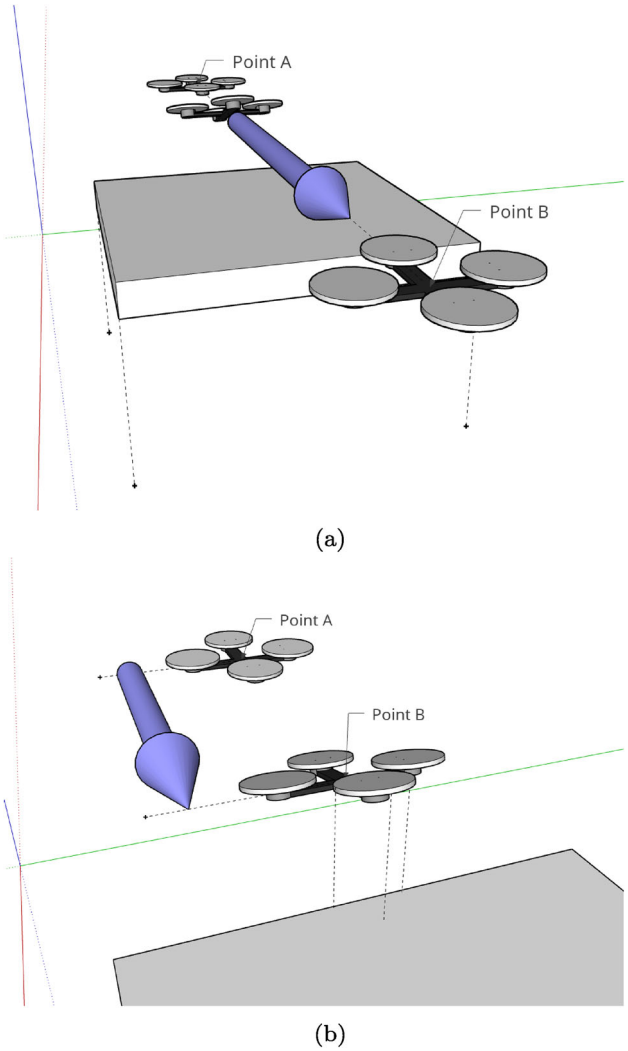


Fig. 14 Two flight cases for testing the influence of the ground in multirotors, **a** flying over an object under the influence of the ground, and **b** hovering at a point with only one rotor under the influence of the ground (partial GE) [96]. When the heading rotor overlaps the surface, it generates more thrust than the other rotors. This effect induces a torque that, in the end, tends to push the quadrotor away from the object

has been modeled similar to full-scale helicopters (Cheeseman and Bennet). To derive this expression, they use potential flow and the method of images, so that the model considers one source located at the geometric center of each rotor and its corresponding image source. Also, they include an additional term with an empirical coefficient, k_b, to account for flow re-circulation and the central body lift it generates. The expression has the following form:

$$\frac{T}{T_\infty} = \Big[1 - (R/4z)^2 - R^2\left(z/\sqrt{(d^2+4z)^3}\right) - (R^2/2)\left(z/\sqrt{(2d^2+4z^2)^3}\right) - 2R^2\left(z/\sqrt{(b^2+4z^2)^3}\right)k_b\Big]^{-1}, \tag{20}$$

where d is the separation from each rotor axis to its adjacent rotor axes and b is the distance between two opposite rotor axes (diagonal). The authors simulated two flight situations involving an object with a flat upmost surface, refer to Fig. 14. In both cases, the lateral axis (y-coordinate) of the drone remained constant, and the setpoint changed along the frontal axis (x-coordinate). Furthermore, they compared the response of three control alternatives: a conventional scheme with cascaded PID controllers, a scheme with a torque disturbance observer in the attitude loop, and a scheme that uses a measurement of the relative distance from each rotor to the planar surface for implementing a feedforward control term in the attitude loop. Little details are provided about the simulation of the GE, but it seems that the effect is accounted for each rotor as a discontinuous event that is triggered when a rotor's center lies over the planar object. By examining the x-coordinate and pitch angle of the drone, the authors found the best response for the feedforward scheme, followed by the one with the disturbance observer. Then, all control alternatives were assessed in a test bench, in which a quadrotor was restricted to only rotate around the lateral axis (pitch angle). In this setup, the partial GE was introduced by placing a large plate under one of the rotors. Again, the results showcased the superior performance of the feedforward scheme over the one with the disturbance observer, and of the latter over the conventional one. Lastly, the same (partial GE) test was reproduced during normal flight only with the standard PID control scheme, showing that the disturbance in the pitch angle was similar to the one observed in the test bench. Nevertheless, the magnitude of the disturbance was greater in normal flight than in the test bench when the plate is retired. Equation (20) was acknowledged in a book chapter about aerodynamic effects in aerial manipulation [97].

Evidence has been reported about how the GE varies for different types of surfaces. Keshavarzian and Danesjou [98] presented a modified model of the GE for all flight types near the ground that considers a surface quality coefficient. The authors claimed that the ground surface plays a significant role in the GE. The maximum value of GE occurs at flat and hard surfaces. On the other hand, it is less over uneven surfaces. The previous claim was confirmed for two cases: flat concrete and grass. They took measurements of differential pressure below a hovering quadrotor, over each surface and at different heights. It was observed that the values measured over flat concrete were higher than the ones over grass. More pressure means less induced velocity, which translates to a greater thrust ratio in GE. Therefore, the GE is greater over flat concrete than over grass. Then, by simplifying the relationship between induced velocity and aircraft speed, and considering Eq. (4), the proposed model is

$$\frac{T}{T_\infty} = \left\{1 - \left(\frac{R}{4z}\right)^2 \left[1 + \left(\frac{1}{k_s} v_a\right)^2\right]^{-1}\right\}^{-1}, \tag{21a}$$

$$v_a = \frac{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}}{\sqrt{-\frac{\dot{x}^2+\dot{y}^2+\dot{z}^2}{2} + \sqrt{\left(\frac{\dot{x}^2+\dot{y}^2+\dot{z}^2}{2}\right)^2 + v_h^4}}}, \tag{21b}$$

where k_s, x, and y are the surface quality coefficient, position of the drone along the inertial x-axis, and position along the inertial y-axis, respectively. Relying on this model, the authors designed a backstepping control scheme for trajectory tracking that considers Eq. (21) in the computations of the control inputs. The control scheme was evaluated in simulations with noise, but no further validation of Eq. (21) was presented.

Shortly after, Kan et al. [99] investigated the behavior of small-scale rotorcrafts in close proximity to the ground both at hover and in forward flight. Performing an extensive experimental study, the authors found that existing models for helicopters cannot fully describe the forward motion of a quadcopter when it operates close to the ground. Consequently, they introduced two new data-driven models for quadrotors operating in GE, both at hover and in forward flight. Two variants were provided for each model: low/medium speed and high speed. The two models are described in Eqs. (22)–(23), in which the low-/medium-speed variants are denoted as "a," and the high-speed variants as "b." The required thrust to hover is denoted as T_h (under ideal conditions it is equal to $mg/4$ for a quadrotor), and the aircraft speed as V.

$$\frac{T}{T_h} = \frac{1 - \frac{3R}{25z}}{1 + \frac{3}{50}\left(\frac{V}{v_h}\right)^3}, \tag{22a}$$

$$\frac{T}{T_h} = \frac{1 - \frac{3R}{25z}}{1 + \frac{3}{50}\left(\frac{V}{v_h}\right)^3} - \frac{29}{250}\left(\frac{V}{v_h}\right)^3. \tag{22b}$$

$$\frac{T}{T_h} = \left(0.104\frac{R}{z} - 0.0952\right)\left(\frac{V}{v_h}\right)^2 - 0.171\frac{R}{z} + 1.02, \tag{23a}$$

$$\frac{T}{T_h} = p_1\left(\frac{V}{v_h}\right)^3 + p_2\left(\frac{V}{v_h}\right)^2 + p_3\frac{V}{v_h} + p_4. \tag{23b}$$

Terms p_1, p_2, p_3, and p_4 are linear functions of R/z, where

$$p_1 = -0.337\frac{R}{z} + 0.161$$
$$p_2 = 0.773\frac{R}{z} - 0.428$$
$$p_3 = -0.350\frac{R}{z} + 0.182$$
$$p_4 = -0.135\frac{R}{z} + 1. \tag{24}$$

It should be mentioned that the models above, which were obtained by fitting data from an AscTec Hummingbird quadrotor (54 cm motor-to-motor), were validated with data collected from a Bitcraze Crazyflie 2.0 (9.2 cm motor-to-motor). The results show that the proposed models are able to predict T/T_∞ for another quadrotor of different size, mass, controller, and trajectories.

Considering the models discussed throughout the paper, we classified the GE models as shown in Table 2.

3.3 Applications

Acknowledging that the GE will influence a multirotor in certain situations can be used to improve the control performance in such situations. Alternatively, this knowledge can be exploited in more specific applications. The authors in [80] took advantage of the GE to perform a blind terrain mapping. The idea is to fly a quadrotor at a constant height and slow forward velocity over the terrain to be mapped while recording the average rotors speed and x-y position. Due to the GE, a relation between this speed and the quadrotor–terrain distance can be determined experimentally. Also, it is assumed that at slow forward velocities, the thrust produced is approximately the thrust required to hover. Therefore, with the recorded data, the height of the terrain underneath the rotorcraft can be estimated along the traveled path. It is important to mention that this approach is limited to estimate distances in which the GE is affecting the rotorcraft. Using an equivalent approach, a sensorless surface detection and estimation procedure is proposed in [10]. Here, the estimated height from the ground is computed as a function of the throttle command of a quadcopter (previously obtained from experimental data). This estimation is used first to detect the ground, and later to land the drone. Two experiments were reported in which a quadcopter is able to land autonomously using this strategy. Nevertheless, the estimation is shown to be reliable only when the height from the ground is below 10 cm, which could lead to a crash. In the same work, ground and ceiling surfaces are considered to achieve energy-efficient path planning. From a heuristic energy relation, the authors demonstrated that it is possible to save about 10% when flying in proximity to horizontal surfaces.

In [100], the author presented a validation procedure, in which the GE was involved, for a stochastic external wrench estimator designed for quadrotors. Basically, the output of the estimator for a quadrotor in GE at different heights was compared to an established model of the GE. It was revealed that

Table 2 Classification of the ground effect models

Model type	References	Remarks	Pros and Cons
Traditional	[14,15,43,78]	Models assuming a single rotor over flat ground	P: computationally efficient, widely validated; C: transients are ignored, best suited for a single rotor
Empirical	[54,63,96]	These models were formulated using empirical data collected with small rotorcraft, either for a single rotor or multiple rotors	P: computationally efficient, multirotors are considered; C: narrowly validated, empirical data are required for fine tuning
Inspired	[59,60,69,80,82]	Models inspired by the ones above considering minor modifications	P: computationally efficient, multirotors are considered; C: narrowly validated, empirical data are required
Parametric	[58,83,86,98,99]	Models considering major modifications to the traditional ones, such as extra parameters related to the surface, rotor geometry, number of rotors, or rotor separation	P: ability to generalize to different situations; C: more involved than previous models, narrowly validated
Learned	[88,89,92,95]	These are machine learning models trained with experimental data collected with small multirotors	P: great performance for a given platform; C: empirical data are required, retraining is required for different platforms
Dynamic	[93,94]	Such models attempt capturing transient or steady-state dynamics that arise when flying near the ground	P: transients are considered; C: narrowly validated

the force estimator provides comparable results to Eq. (2). Another application consisted of using the GE to provide passive lateral position stability for a quadrotor with canted rotors flying in a sloped ground channel [101]. By testing a variety of rotor cant and channel slope angles, a combination was found that gave the best balance between settling time and overshoot for the passive lateral position control task.

4 Other proximity effects

Small rotorcrafts are well suited to carrying out missions within confined environments. Operation in such environments will place a small rotorcraft in close proximity to surfaces, not only to the ground. The second most mentioned proximity interaction is the Ceiling Effect (CE) [54,79,82,87,97], which emerges when a rotorcraft is approaching from below to a horizontal surface. Sanchez-Cuevas et al. [102] proposed a solution for bridge inspection that uses the CE to maintain contact with a specially designed quadcopter. Wang et al. [103] investigated the aerodynamic performance of a quadrotor hovering under the ceiling. They formulated a model of the CE by introducing a correction coefficient into Eq. (2) to replace the 4. After that, they performed a series of CFD simulations to get the data to estimate the coefficient. The fitted model indicates that the CE still exists at $z/R = 2$. Kocer et al. [104] presented a cascaded control structure for quadrotors subjected to the CE. The control structure consists of an attitude controller (PID) and a position controller (nonlinear model predictive). Experimental results demonstrated the effectiveness of the position controller.

Besides horizontal surfaces, vertical surfaces or walls also alter the flow through a rotorcraft. For a small hovering rotor, the authors in [105] addressed the (single) wall effect (WE) with CFD simulations. The results showed that the wall induces an important wake asymmetry, which produces disturbance forces and moments acting on the rotor disk. As an extension [106], the authors considered the impact of small variations in rotor attitude on airflow–wall interactions. It was found that, for a blade near a wall, the lift force fluctuates periodically as a function of the blade azimuth angle. These fluctuating lift forces generate moments at the rotor disk that vary as a function of rotor–wall separation and attitude. The WE is also contemplated in [54,87,97,100].

The introduction of the ground, ceiling, and wall effects to the literature has incentived the emergence of combinations. For example, the authors in [107], whose work has been already described in Sect. 3.1, evaluated their adaptive control proposal with a quadrotor that was required to hover near the ground and two walls. On the other hand, the authors in [56,57] displayed three extra configurations: channel (ground and ceiling), low corner (ground and wall), and upper corner (ceiling and wall). They studied the lower corner effect numerically and experimentally for a hovering rotor. A different situation was first depicted in [97], which the authors called the pipe effect. This effect does not only depend on the distance to the obstacle, but also the size of

the rotor and the tube. The pipe ground and ceiling effects were experimentally explored in [82].

5 Discussion

With the study of aerodynamic proximity effects on rotorcrafts, many types of problems appear when simulations and experiments are carried out. To solve them, researchers around the globe have presented new and exciting ideas. With the ongoing development of research, various trends have been formed. However, there remain many challenges on how to deal with aerodynamic proximity effects. Some of these have been addressed, whereas others remain open, refer to Fig. 2.

The autonomous navigation of rotorcraft in near-surface conditions demands better sensing and algorithms. Even though it is important to accomplish onboard perception and data processing, UAVs impose restrictions on the weight of the hardware, and therefore, the processing speed onboard the vehicle. The use of MoCap systems limits the application scene, and it is slowly being replaced by onboard vision cameras [74,108]. As processing technology advances, onboard data processing will be enabled, allowing to dispense with the wireless data transmission segment. Without this segment, the real-time computing capabilities of a UAV increase. Summing up, two major challenges remain to be solved for the perception, localization, and processing problem: fusion and filtering of diversified sensors, and real-time onboard data processing.

In addition, an increasing number of researchers are adopting the parametric modeling approach [87,99]. Different from classical (traditional) [14,15] and empirical approaches [63,96], the parametric approach attempts to provide models that consist of parametrized equations that can be applied to a variety of multirotor designs. These parameters can include variables such as rotor diameter, rotor spacing, number of rotors, just to mention a few. On the other hand, the dynamic approach tries to model the transients that occur when a rotorcraft flies close to a surface [93,94]. The empirical approach faces the challenge of experimentally identifying any coefficient, which will be valid only for a single rotorcraft. For its part, the dynamic approach should solve the decoupling between proximity effects and control action. Moreover, all new models should endure further validation.

Meanwhile, controllers have been developed from the early PID to the present situation, in which adaptive [63], robust [98], backstepping [48], and feedback plus feedforward [12,96] are used. A controller should address the following challenges: handle sensor faults that are common for current sensors [74], and exploit new models of aerodynamic interactions, such as those in [87,99]. In terms of applications, the literature has moved from merely improving the control performance to high-level tasks, such as passive proximity estimation [54], and energy-efficient navigation [10]. In this sense, a major challenge is posed: *autonomous navigation in unknown confined environments.* We argue that the latter should not be studied from the point of view of control theory only, but it also has to involve other approaches whose basis are found in modern artificial intelligence techniques such as those based on computer vision and robotics. For instance, nowadays robotic systems rely on well-known techniques such as visual SLAM for localization and representation of the environment [74], with the caveat that their operation frequency runs much slower than MoCap systems (< 60 Hz). However, even this limitation is an opportunity to develop mechanisms to mitigate or compensate for the GE under low-frequency observations.

6 Conclusion

We have presented a literature review on the study of the ground effect for small-scale rotorcraft UAVs. First, we provided a summary of the origins of this topic in full-scale helicopters. We proceeded to discuss the works that naturally extended the topic to model helicopters. After that, we reviewed the body of research that investigate the influence of the ground on small multirotor vehicles. We also provided a brief section describing other aerodynamic proximity effects, such as the ceiling and wall effects. Finally, we included a discussion section where problems, trends, and challenges were disclosed.

In general, the subjects of study in this review were threefold: control, modeling, and applications. Four main control strategies were identified: adaptive, observer/estimator based, robust, and fault-tolerant. On the other hand, modeling studies can be classified into three groups: classical/empirical, parametric, and dynamic. Also, it was noticed that acknowledging proximity effects has the following general applications: control improvement, passive proximity estimation, and energy-efficient navigation.

This review shows that research on aerodynamic proximity effects is rapidly expanding and growing. It is also evident that a lot of work remains to be undertaken in the future. This includes extensive validation of new models, application of new models in control architectures, and autonomous navigation in unknown confined environments, where we argue that artificial intelligence techniques based on computer vision and robotics may leverage the emergence of novel solutions. With the advance of simulation software, we envisage that computational fluid dynamics will be used to identify empirical models of proximity effects offline while assuring a high level of accuracy. From the point of view of biomimetics, we envisage that onboard multicamera systems in tandem

with artificial intelligence will be adopted for sensing and controlling aerodynamic proximity effects. Regarding the multicamera systems, we foresee that high-speed and event cameras may prove useful [109].

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References

1. Naidoo Y, Stopforth R, Bright G (2011) Development of an UAV for search & rescue applications. In: IEEE Africon'11, Livingstone, Zambia, September 2011, pp 1–6
2. Irizarry J, Gheisari M, Walker BN (2012) Usability assessment of drone technology as safety inspection tools. J Inf Technol Constr 17:194–212
3. Natalizio E, Surace R, Loscri V, Guerriero F, Melodia T (2012) Filming sport events with mobile camera drones: mathematical modeling and algorithms. Technical report HAL-00801126, HAL-Inria, December
4. Momont A (2014) Ambulance drone. https://www.tudelft.nl/en/ide/research/research-labs/applied-labs/ambulance-drone/. Accessed 29 Feb 2020
5. Muoio D (2017) Uber partners with aurora flight sciences on flying taxis—business insider. https://www.businessinsider.com/uber-partners-with-aurora-flight-sciences-on-flying-taxis-2017-4. Accessed 29 Feb 2020
6. Gallo MF (2013) Aviation accident final report CEN11FA507 (fatal). Accident report, National Transportation Safety Board, USA, February, p 2013
7. Rayner BC (2016) Aviation accident final report ERA16CA160 (non-fatal). Accident report, National Transportation Safety Board, USA, June
8. Mahony R, Hamel T (2001) Adaptive compensation of aerodynamic effects during takeoff and landing manoeuvres for a scale model autonomous helicopter. Eur J Control 7(1):43–57
9. Guenard N, Hamel T, Eck L (2006) Control laws for the tele operation of an unmanned aerial vehicle known as an X4-flyer. In: 2006 IEEE/RSJ international conference on intelligent robots and systems, Beijing, China, October 2006, pp 3249–3254
10. Gao S, Di Franco C, Carter D, Quinn D, Bezzo N (2019) Exploiting ground and ceiling effects on autonomous UAV motion planning. In: 2019 international conference on unmanned aircraft systems (ICUAS), Atlanta, GA, USA, June, pp 768–777
11. Kenichiro N, Hirokazu S (2011) Integral sliding mode altitude control for a small model helicopter with ground effect compensation. In: Proceedings of the 2011 American control conference, San Francisco, CA, USA, June 2011. IEEE, pp 202–207
12. John B, Andrew C, Walterio M-C (2015) Improving MAV control by predicting aerodynamic effects of obstacles. In: 2015 IEEE/RSJ international conference on intelligent robots and systems (IROS), Hamburg, Germany, September 2015, pp 4826–4833
13. Albert B (1937) The ground effect on lifting propellers. Technical report 835, National Advisory Committee for Aeronautics, Washington, DC, USA
14. Cheeseman IC, Bennett WE (1955) The effect of the ground on a helicopter rotor in forward flight. Technical report 3021, Aeronautical Research Council, London, UK, September
15. Hayden JS (1976) The effect of the ground on helicopter hovering power required. In: Proceeding of AHS 32nd annual forum, Washington, DC, USA, May
16. Fradenburgh E (1960) The helicopter and the ground effect machine. J Am Helicopter Soc 5
17. Marr RL, Ford DG, Ferguson SW (1974) Analysis of the wind tunnel test of a tilt rotor powered force model. Technical report NASA-CR-137529, National Aeronautics and Space Administration, Moffett Field, CA, USA, June
18. Cai Guowei, Dias Jorge, Seneviratne Lakmal (2014) A survey of small-scale unmanned aerial vehicles: recent advances and future development trends. Unmanned Syst 2(2):1–25
19. Leishman JG (2006) Principles of helicopter aerodynamics, 2nd edn. Cambridge University Press, New York
20. Johnson Wayne (1994) Helicopter theory, revised edn. Dover Publications Inc, New York
21. Prouty Raymond W (2001) Helicopter performance, stability, and control, 2nd edn. Krieger Publishing Company, Malabar
22. Knight M, Hegner RA (1941) Analysis of ground effect on the lifting airscrew. Technical report 835, National Advisory Committee for Aeronautics, Washington, DC, USA, December
23. Zbrozek JK (1947) Ground effect on the lifting rotor. Technical report 2347, Aeronautical Research Council, London, UK, July
24. Light JS (1989) Tip vortex geometry of a hovering helicopter rotor in ground effect. In: 45th AHS annual forum, Boston, MA, USA, May
25. Griffiths DA, Ananthan S, Leishman JG (2005) Predictions of rotor performance in ground effect using a free-vortex wake model. J Am Helicopter Soc 50(4):302–314
26. Pulla D (2006) A study of helicopter aerodynamics in ground effect. PhD thesis, Ohio State University, Columbus, OH, USA
27. Filippone A, Bakker R, Basset PM, Rodriguez B, Green R, Bensing F, Visingardi A (2011) Rotor wake modelling in ground effect conditions. In: 37th European rotorcraft forum, Vergiate and Gallarate, Italy, September 2011
28. Lakshminarayan Vinod K, Kalra Tarandeep S, Baeder James D (2013) Detailed computational investigation of a hovering microscale rotor in ground effect. AIAA J 51:893–909
29. Kalra TS (2014) CFD modeling and analysis of rotor wake in hover interacting with a ground plane. PhD thesis, University of Maryland, College Park, MD, USA
30. Brown Richard E, Whitehouse Glen R (2004) Modelling rotor wakes in ground effect. J Am Helicopter Soc 49(3):238–249
31. Lee T, Leishman JG, Ramasamy M (2010) Fluid dynamics of interacting blade tip vortices with a ground plane. J Am Helicopter Soc 55:022005
32. Desopper A, Routhieau V, Roth G, von Grunhagen W, Haverdings H (2002) Study of the low speed characteristics of a tiltrotor. In: 28th European rotorcraft forum, Bristol, UK, September 2002
33. Felker FF (1988) A review of tilt rotor download research. In: 14th European rotorcraft forum, Milan, Italy, September 1988
34. Lestari A, Niazi S, Rajagopalan RG (2001) Preliminary numerical analysis of a quad tilt rotor flowfield and performance. In: Tiltrotor/runway independent aircraft technology and applications

specialists'. Meeting of the American Helicopter Society, Arlington, TX, USA, March 2001
35. Radhakrishnan A, Schmitz F (2003) An experimental investigation of a quad tilt rotor in ground effect. In: 21st AIAA applied aerodynamics conference, Orlando, FL, USA, June 2003
36. Gupta V, Baeder JD (2002) Quad tiltrotor aerodynamics in ground effect. In: 58th annual forum of the American helicopter society, Montreal, QC, Canada, June 2002
37. Heyson HH (1960) Ground effect for lifting rotors in forward flight. Technical report NASA-TN-D-234, National Aeronautics and Space Administration, Hampton, VA, USA, May 1960
38. Curtiss HC, Sun M, Putman WF, Hanker EJ (1984) Rotor aerodynamics in ground effect at low advance ratios. J Am Helicopter Soc 29(1):48–55
39. Ganesh B, Komerath N (2004) Unsteady aerodynamics of rotorcraft in ground effect. In: 22nd applied aerodynamics conference and exhibit, Reno, NV, USA, January 2004
40. Nathan ND (2010) The rotor wake in ground effect and its investigation in a wind tunnel. PhD thesis, University of Glasgow, Scotland, UK
41. Nathan ND, Green RB (2012) The flow around a model helicopter main rotor in ground effect. Exp Fluids 52(1):151–166
42. Heyson HH (1977) Theoretical study of the effect of ground proximity on the induced efficiency of helicopter rotors. Technical report TM-X-71951, National Aeronautics and Space Administration, Hampton, VA, USA, May 1977
43. Khromov V, Rand O (2008) Ground effect modeling for rotary-wing simulation. In: 26th international congress of the aeronautical sciences, Anchorage, AK, USA, September 2008, pp 14–19
44. Bludau J, Rauleder J, Friedmann L, Hajek M (2016) Real-time simulation of rotor inflow using a coupled flight dynamics and fluid dynamics simulation. In: Deutscher Luft- und Raumfahrtkongress 2016, Braunschweig, Germany, September 2016
45. Yu Z, Kenzo N, Jinok S, Demian C (2007) 3D vision based landing control of a small scale autonomous helicopter. Int J Adv Robot Syst 4(1):51–56
46. Cesetti A, Frontoni E, Mancini A, Zingaretti P, Longhi S (2010) A vision-based guidance system for uav navigation and safe landing using natural landmarks. J Intell Robot Syst 57(1):233–257
47. Roy TK, Garratt M, Pota HR, Samal MK (2012) Robust altitude control for a small helicopter by considering the ground effect compensation. In: Proceedings of the 10th world congress on intelligent control and automation, Beijing, China, July 2012, pp 1796–1800
48. Roy TK, Suman AA (2013) Adaptive backstepping controller for altitude control of a small scale helicopter by considering the ground effect compensation. In: 2013 international conference on informatics, electronics and vision (ICIEV), Dhaka, Bangladesh, May 2013, pp 1–5
49. Roy Tushar K (2013) Robust adaptive backstepping controller for altitude control of a small scale helicopter by considering the ground effect compensation. Univ J Control Autom 1(2):52–57
50. Hooi CG (2015) Height estimation and control of a rotorcraft in ground effect using multiple pressure probes. Master's thesis, University of Maryland, College Park, MD, USA, 2015
51. Hooi CG, Lagor FD, Paley DA (2015) Flow sensing, estimation and control for rotorcraft in ground effect. In: 2015 IEEE aerospace conference, Big Sky, MT, USA, March 2015, pp 1–8
52. Hooi CG, Lagor FD, Paley DA (2015) Flow sensing for height estimation and control of a rotor in ground effect: modeling and experimental results. In: AHS 71st annual forum, Virginia Beach, VA, USA, May 2015, pp. 1–12
53. Hooi CG, Lagor FD, Paley DA (2016) Height estimation and control of rotorcraft in ground effect using spatially distributed pressure sensing. J Am Helicopter Soc 61(4):1–14
54. Robinson DC (2016) Modelling and estimation of aerodynamic disturbances acting on a hovering micro helicopter in close proximity to planar surfaces. PhD thesis, Monash University, Melbourne, Australia
55. Jardin T, Prothin S, Magaña CG (2017) Aerodynamic performance of a hovering microrotor in confined environment. J Am Helicopter Soc 62(2):1–7
56. Prothin S, Escudero CF, Jardin T, Doue N (2018) ARChEaN: aerodynamics of rotors in confined environments study in ground and corner effect. In: 10th international micro-air vehicles conference, Melbourne, Australia, November 2018
57. Prothin S, Escudero CF, Doue N, Jardin T (2019) Aerodynamics of MAV rotors in ground and corner effect. Int J Micro Air Veh 11
58. Hosseini Zahra, Ramirez-Serrano Alejandro, Martinuzzi Robert J (2011) Ground/wall effects on a tilting ducted fan. Int J Micro Air Veh 3(3):119–141
59. He X, Leang KK (2019) A new quasi-steady in-ground effect model for rotorcraft unmanned aerial vehicles. In: ASME, dynamic systems and control conference, vol 3, Park City, UT, USA, October
60. Lee D, Kim HJ, Sastry S (2009) Feedback linearization vs. adaptive sliding mode control for a quadrotor helicopter. Int J Control Autom Syst 7(3):419–428
61. Hu B, Lu L, Mishra S (2015) Fast, safe and precise landing of a quadrotor on an oscillating platform. In: 2015 American control conference (ACC), Chicago, IL, USA, July 2015, pp 3836–3841
62. Arshad U (2019) Adaptive control implementation to include ground effect on UAV simulator. Master's thesis, Politecnico di Milano, Milan, Italy
63. Wei P, Chan SN, Lee S, Kong Z (2019) Mitigating ground effect on mini quadcopters with model reference adaptive control. Int J Intell Robot Appl 3(3):283–297
64. Nobahari H, Sharifi AR (2014) Continuous ant colony filter applied to online estimation and compensation of ground effect in automatic landing of quadrotor. Eng Appl Artif Intell 32:100–111
65. Du H, Pu Z, Jianqiang Y, Hanbo Q (2016) Advanced quadrotor takeoff control based on incremental nonlinear dynamic inversion and integral extended state observer. In: 2016 IEEE Chinese guidance. Navigation and control conference (CGNCC). Nanjing, China, pp 1881–1886
66. He X, Kou G, Calaf M, Leang KK (2019) In-ground-effect modeling and nonlinear-disturbance observer for multirotor unmanned aerial vehicle control. J Dyn Syst Meas Control 141(7)
67. He X, Calaf M, Leang KK (2017) Modeling and adaptive nonlinear disturbance observer for closed-loop control of in-ground-effects on multi-rotor UAVs. In: ASME, dynamic systems and control conference. Tysons Corner, VA, USA, October
68. Waheed S, Awan AU, Khan KFA, Liaquat M (2015) Robust tracking of quadrotor UAV using RISE controller with input saturation. In: 2015 34th Chinese control conference (CCC), Hangzhou, China, July 2015, pp 5507–5512
69. Danjun L, Yan Z, Zongying S, Geng L (2015) Autonomous landing of quadrotor based on ground effect modelling. In: 2015 34th Chinese control conference (CCC), Hangzhou, China, July 2015, pp. 5647–5652
70. Shi G, Shi X, O'Connell M, Yu R, Azizzadenesheli K, Anandkumar A, Yue Y, Chung S (2019) Neural lander: stable drone landing control using learned dynamics. In: 2019 international conference on robotics and automation (ICRA), Montreal, QC, Canada, May 2019, pp 9784–9790
71. From H (2019) Predicting the ground effect in drone landing with online learning. Master's thesis, KTH Royal Institute of Technology, Stockholm, Sweden

72. Herissé B, Hamel T, Mahony R, Russotto F (2012) Landing a VTOL unmanned aerial vehicle on a moving platform using optical flow. IEEE Trans Robot 28(1):77–89
73. Lee D, Ryan T, Kim HJ (2012) Autonomous landing of a VTOL UAV on a moving platform using image-based visual servoing. In: 2012 IEEE international conference on robotics and automation, Saint Paul, MN, USA, May 2012, pp 971–976
74. Matus-Vargas Antonio, Rodriguez-Gomez Gustavo, Martinez-Carranza Jose (2019) A monocular SLAM-based controller for multirotors with sensor faults under ground effect. Sensors 19(22):4948
75. Nakanishi H, Kanata S, Sawaragi T (2011) Measurement model of barometer in ground effect of unmanned helicopter and its application to estimate terrain clearance. In: 2011 IEEE international symposium on safety, security, and rescue robotics, Kyoto, Japan, November 2011. IEEE
76. Matus-Vargas A, Rodriguez-Gomez G, Martinez-Carranza J (2019) Aerodynamic disturbance rejection acting on a quadcopter near ground. In: 2019 6th international conference on control. decision and information technologies (CoDIT), France, Paris, pp 1516–1521
77. Matus-Vargas A, Rodriguez-Gomez G, Martinez-Carranza J (2019) Sensor fault mitigation for MAVs under ground effect. In: 2019 workshop on research, education and development of unmanned aerial systems (RED UAS). Cranfield, UK, November
78. Bangura M, Mahony R (2012) Nonlinear dynamic modeling for high performance control of a quadrotor. In: Proceedings Australasian conference on robotics and automation 2012, Wellington, New Zealand, October 2012. Australian Robotics and Automation Association, pp 1–10
79. Belatti T, Powers C, Kumar V (2012) Quadrotor flight in constrained environments. Technical report, University of Pennsylvania
80. Powers C, Mellinger D, Kushleyev A, Kothmann B, Kumar V (2013) Influence of aerodynamics and proximity effects in quadrotor flight. In: The 13th international symposium on experimental robotics, Quebec, QC, Canada, June 2013. Springer International Publishing, pp 289–302
81. Sharf I, Nahon M, Harmat A, Khan W, Michini M, Speal N, Trentini M, Tsadok T, Wang T (2014) Ground effect experiments and model validation with Draganflyer X8 rotorcraft. In: 2014 international conference on unmanned aircraft systems (ICUAS), Orlando, FL, USA, May 2014, pp 1158–1166
82. Sanchez-Cuevas PJ, Martin V, Heredia G, Ollero A (2019) Aerodynamic effects in multirotors flying close to obstacles: modelling and mapping. In: Robot 2019: fourth iberian robotics conference, advances in intelligent systems and computing, Porto, Portugal, November 2019. Springer International Publishing, pp 63–74
83. Eberhart GM (2017) Modeling of ground effect benefits for multirotor small unmanned aerial systems at hover. Master's thesis, Ohio University, Athens, OH, USA
84. Sanchez-Cuevas PJ, Heredia G, Ollero A (2017) Experimental approach to the aerodynamic effects produced in multirotors flying close to obstacles. In: ROBOT 2017: third Iberian robotics conference, advances in intelligent systems and computing, Seville, Spain, November 2017. Springer International Publishing, pp 742–752
85. Bernard DDC, Giurato M, Riccardi F, Lovera M (2017) Ground effect analysis for a quadrotor platform. In: Fourth CEAS specialist conference on guidance, navigation and control, advances in aerospace guidance, navigation and control, Warsaw, Poland, April 2017. Springer International Publishing, pp 351–367
86. Conyers SA, Rutherford MJ, Valavanis KP (2018) An empirical evaluation of ground effect for small-scale rotorcraft. In: 2018 IEEE international conference on robotics and automation (ICRA), Brisbane, QLD, Australia, May 2018, pp 1244–1250
87. Conyers SA (2019) Empirical evaluation of ground, ceiling, and wall effect for small-scale rotorcraft. PhD thesis, University of Denver, Denver, CO, USA
88. Ryan T, Kim HJ (2012) Modelling of quadrotor ground effect forces via simple visual feedback and support vector regression. In: AIAA guidance, navigation, and control conference, guidance, navigation, and control and co-located conferences, Minneapolis, MN, USA, August 2012. American Institute of Aeronautics and Astronautics
89. Mohajerin N, Waslander SL (2015) Modelling a quadrotor vehicle using a modular deep recurrent neural network. In: 2015 IEEE international conference on systems. Man, and Cybernetics. Kowloon, China, pp 376–381
90. Karydis Konstantinos, Kumar Vijay (2017) Energetics in robotic flight at small scales. Interface Focus 7(1):20160088
91. Karydis K, Hsieh MA (2016) Uncertainty quantification for small robots using principal orthogonal decomposition. In: 2016 international symposium on experimental robotics, Tokyo, Japan, October 2016. Springer International Publishing, pp 33–42
92. Karydis Konstantinos, Poulakakis Ioannis, Sun Jianxin, Tanner Herbert G (2015) Probabilistically valid stochastic extensions of deterministic models for systems with uncertainty. Int J Robot Res 34(10):1278–1295
93. Aich S, Ahuja C, Gupta T, Arulmozhivarman P (2014) Analysis of ground effect on multi-rotors. In: 2014 international conference on electronics, communication and computational engineering (ICECCE), Hosur, India, November 2014. IEEE, pp 236–241
94. Del CD, Bernard FR, Giurato M, Lovera M (2017) A dynamic analysis of ground effect for a quadrotor platform. In: 20th IFAC world congress, vol 50. Tolouse, France, pp 10311–10316
95. Bartholomew J, Calway A, Mayol-Cuevas W (2014) Learning to predict obstacle aerodynamics from depth images for micro air vehicles. In: 2014 IEEE international conference on robotics and automation (ICRA), Hong Kong, China, May 2014, pp 4967–4973
96. Sanchez-Cuevas Pedro J, Heredia Guillermo, Ollero Anibal (2017) Characterization of the aerodynamic ground effect and its influence in multirotor control. Int J Aerosp Eng 2017:17
97. Sanchez-Cuevas PJ, Heredia G, Ollero A (2019) Multirotor Aerodynamic effects in aerial manipulation. Springer International Publishing, Cham, Switzerland, 2019, pp 67–82
98. Keshavarzian Hasan, Daneshjou Kamran (2019) Modified underactuated quadrotor model for forwarding flight in the presence of ground effect. Aerosp Sci Technol 89:242–252
99. Kan Xinyue, Thomas Justin, Teng Hanzhe, Tanner Herbert G, Kumar Vijay, Karydis Konstantinos (2019) Analysis of ground effect for small-scale UAVs in forward flight. IEEE Robot Autom Lett 4(4):3860–3867
100. McKinnon CD (2015) Data driven, force based interaction for quadrotors. Master's thesis, University of Toronto, Toronto, ON, Canada
101. Davis E, Pounds PEI (2016) Passive position control of a quadrotor with ground effect interaction. IEEE Robot Autom Lett 1(1):539–545
102. Sanchez-Cuevas PJ, Heredia G, Ollero A (2017) Multirotor uas for bridge inspection by contact using the ceiling effect. In: 2017 international conference on unmanned aircraft systems (ICUAS), Miami, FL, USA, June 2017, pp 767–774
103. Wang X, Du S, Yong L (2017) Research on ceiling effect of quadrotor. In: 2017 IEEE 7th annual international conference on CYBER technology in automation. Control, and intelligent systems (CYBER). Honolulu, HI, USA, pp 846–851
104. Kocer BB, Tjahjowidodo T, Seet GGL (2018) Centralized predictive ceiling interaction control of quadrotor VTOL UAV. Aerosp Sci Technol 76:455–465
105. Robinson DC, Chung H, Ryan K (2014) Computational investigation of micro rotorcraft near-wall hovering aerodynamics. In: 2014

international conference on unmanned aircraft systems (ICUAS), Orlando, FL, USA, May 2014, pp 1055–1063
106. Robinson DC, Chung H, Ryan K (2014) Computational investigation of micro helicopter near-wall effect. In: 19th Australasian fluid mechanics conference (AFMC), Melbourne, Australia, December 2014. RMIT University
107. Lee D, Awan A, Kim S, Kim HJ (2012) Adaptive control for a VTOL UAV operating near a wall. In: AIAA guidance, navigation, and control conference, Minneapolis, MN, USA, August 2012
108. Moon H, Martinez-Carranza J, Cieslewski T, Faessler M, Falanga D, Simovic A, Scaramuzza D, Li S, Ozo M, De Wagter C, de Croon G, Hwang S, Jung S, Shim H, Kim H, Park M, Au T-C, Kim SJ (2019) Challenges and implemented technologies used in autonomous drone racing. Intell Serv Robot 12(2):137–148
109. Rebecq H, Ranftl R, Koltun V, Scaramuzza D (2019) High speed and high dynamic range video with an event camera. IEEE Trans Pattern Anal Mach Intell

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